

Continuous functions – 9

Approximation: Weierstrass, Bernstein, Stone

1 сентября 2026 года

Theoretical minimum.

Theorem 1 (Weierstrass). Every continuous function on $[a, b]$ is a uniform limit of polynomials; the same holds for continuous 2π -periodic functions and trigonometric polynomials.

Theorem 2 (Bernstein). For $B_n f(x) = \sum_{k=0}^n f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k}$ one has $\|B_n f - f\|_\infty \leq \frac{3}{2} \omega_f(n^{-1/2})$, which proves Theorem 1 constructively and with an explicit rate.

Theorem 3 (Stone–Weierstrass). A subalgebra of $C(K)$ on a compact K which contains the constants and separates points is dense. Hence polynomials in x^2 are dense in the even functions, and the algebra generated by $\sin x$ and $\cos x$ is dense in the periodic ones.

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1. Prove Theorem 2 (the probabilistic proof through the law of large numbers is allowed) and deduce Weierstrass' theorem with the explicit estimate through the modulus of continuity.
 2. Prove that if f is continuous on $[0, 1]$ and $\int_0^1 f(x) x^n dx = 0$ for every $n \geq 0$, then $f \equiv 0$. What changes if the equalities are assumed only for even n ?
 3. Prove the Stone–Weierstrass theorem in the version for a compact metric space, and deduce that the polynomials in x^2 are dense in the space of even continuous functions on $[-1, 1]$, while they are not dense in $C[-1, 1]$.
 4. Prove that every continuous function on $[a, b]$ is a uniform limit of piecewise-linear functions, and estimate the number of pieces needed in terms of the modulus of continuity.

5. State Müntz's theorem: the span of $\{x^{\lambda_n}\}$ with $\lambda_n \rightarrow \infty$ is dense in $C[0, 1]$ if and only if $\sum \frac{1}{\lambda_n} = \infty$. Verify the necessity on the example $\lambda_n = 2^n$ using the orthogonality argument.